

A recap of what we did in class, plus the standard way economists classify markets, which the textbook does not spell out. Slide numbers refer to the Lecture 12 deck. Reading: CORE 7.8, 7.9, 7.11, 7.12, and the markup rule in 7.6.

1. Market Power and the Markup Rule

A firm has **market power** if it can price above marginal cost without losing all of its buyers to rivals. Every firm since Cheerios has had some: each chose the quantity where $MR = MC$ and charged a price above MC. How far above? The **profit margin** is $P - MC$; the **price markup** is that margin as a share of the price. A profit-maximizing firm's markup equals one over the price elasticity of its demand.

The markup rule

$$\frac{P - MC}{P} = \frac{1}{\varepsilon}$$

The less elastic a firm's demand, the higher its markup.

Check it on the quiz's scooter company: $P = 800$ and $MC = 400$, so the markup is $1/2$; the elasticity there is $\varepsilon = (800/20) \times (1/20) = 2$, and $1/\varepsilon = 1/2$. Demand is less elastic when buyers have few close **substitutes**. In the car data (slide 6), small cars with many rivals had elasticities near 6 and markups of 16%; luxury cars with few rivals had elasticities near 3 and markups near 30%. When a drug's patent expires, each generic maker is a perfect substitute for the others, and the price falls to 61% of the brand's with one generic maker and 6% with six (slide 7).

2. Two Sources of Market Power

Product differentiation. The firm sells something a little different from its rivals, so its buyers do not all leave when it charges a little more. Firms differentiate through location and service, through innovation (the Prius had few close rivals for about fifteen years), and through advertising, which informs buyers, draws them from rivals, and builds brand loyalty. In the cereal market, advertising raised a brand's demand more than price cuts did.

Cost advantages of size. In some industries average cost falls as output grows, because a large fixed cost is spread over more units. A movie costs \$100 million to make and \$1 to stream to one more viewer, so AC is \$21 a viewer at 5 million viewers, \$11 at 10 million, \$5 at 25 million, and \$3 at 50 million (slide 10). Since profit is $(P - AC) \times Q$, a firm **breaks even** where price equals average cost and loses money wherever AC is above the price. At a \$5 price a studio needs 25 million viewers; with fewer it loses on every stream, and beyond that its profit grows with its audience.

This is a different question from Lecture 9's. There, $MR = MC$ picked the best quantity and price, and a fixed cost did not change them. But the best choice is not always profitable: if the price is below AC at that quantity, even the best choice loses money. So $MR = MC$ answers *how much to produce*, and price against AC answers *whether producing pays at all*.

That second test is why falling average costs limit competition. Two studios splitting the 50 million viewers each just break even; three would all lose. Each pays the whole fixed cost for a share of the buyers, so only a few large firms survive and buyers have few sellers to choose from. In the

extreme, one firm can supply the whole market at a lower AC than two could: a **natural monopoly**, as with water, electricity, and gas networks, where the pipes and wires are the fixed cost. When AC falls steeply enough, competition becomes winner-takes-all, as in internet search.

3. The Four Market Structures

CORE describes a spectrum from **price-takers**, too small to affect the going price, to **price-setters** with real power over theirs. Most textbooks, and your later courses, divide it into four **market structures** by how many firms sell, whether products are identical or differentiated, and how hard entry is.

The four market structures

Structure	Firms	Products	Entry	Control over price	Example
Perfect competition	Many	Identical	Easy	None	Wheat farms
Monopolistic competition	Many	Differentiated	Easy	Some	Restaurants
Oligopoly	Few	Either	Hard	Some, rivals react	Wireless carriers
Monopoly	One	No substitutes	Blocked	The most	Water utility

Perfect competition. Many sellers, an identical product, free entry. No seller can charge above the going price, since buyers would switch to one of the many others. Each firm is a price-taker with perfectly elastic demand: markup zero, price equal to marginal cost. Wheat farmers are the usual example; CORE develops this case in Unit 8, after the midterm.

Monopolistic competition. Many sellers and easy entry, but each product differs a little by location, style, or brand: restaurants, hair salons, clothing stores. Each firm faces its own downward-sloping demand and prices above MC, but close substitutes keep demand fairly elastic, and any profit attracts entrants. Most firms in Unit 7, from Cheerios to Beautiful Cars, are of this kind.

Oligopoly. A few large firms, hard entry (usually from cost advantages of size), identical or differentiated products. What is distinctive is that each firm's best price depends on what its rivals charge: a carrier that cuts its price takes customers from the others, who must decide whether to follow. The next lecture models this as a game.

Monopoly. One seller of a product with no close substitutes, protected by its cost structure, a patent, control of a key resource, or a legal grant. It faces the whole market's demand, the least elastic a firm can face, so its markup is the largest. Few firms face no competition at all; how much power a seller has depends on how widely the market is drawn.

4. Competition Policy

In every structure but perfect competition the price is above MC, so some buyers who value the good above its cost do not get it: the outcome is not Pareto efficient and there is a deadweight loss (Lecture 11). Estimates for US firms put the average price 21% above MC in 1980 and 61% above in 2016 (slide 14).

Competition policy, or **antitrust policy**, is government policy and law that limits market power, prevents cartels, and regulates how firms compete. Authorities step in when a dominant firm has both the ability and the incentive to abuse its power. What they watch for follows the markup

rule: anything that makes a firm's demand less elastic. Firms can keep rivals out, from land (supermarkets buying sites and leaving them empty), inputs (Ferrero buying a hazelnut supplier), or buyers (Microsoft keeping rival browsers off Windows); merge with rivals (Meta buying Instagram and WhatsApp); or collude in a **cartel**, agreeing to act like one monopoly, which is illegal in most countries but hard to prove. They can also make buyers less price-sensitive through brands, shelf placement, and products that work best together. A **patent** is the one monopoly the law grants on purpose, for a limited time, as a reward for research.

Authorities weigh harms against benefits case by case. A natural monopoly is cheaper to regulate, or own publicly, than to break up. Platforms are a dilemma: scale and network effects make them look like natural monopolies even as many sellers compete on them. And policy plays catch-up: the browser market the Microsoft case opened is now dominated by Chrome.

What to take away

- A firm has market power when buyers do not switch easily to rivals. Its markup is $1/\varepsilon$: the less elastic its demand, the more it charges above marginal cost.
- Market power comes from a product that differs from rivals', or from cost advantages of size that leave room for only a few large firms.
- Markets range from perfect competition through monopolistic competition and oligopoly to monopoly, as firms get fewer, products more distinct, and entry harder.
- A price above marginal cost means a deadweight loss. Competition policy limits market power, except where a natural monopoly is cheaper to regulate than to break up.